

Geospatial Determinants of Game Outcomes in NCAA Men's Basketball

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Abstract: This study examines how travel through space and time affects the results (odds of winning) of 3,296 team performances from all 1,648 NCAA Division I Men's Basketball Tournament games played over 26 years, from 1985/86 to 2010/11. Tournament schools and game sites over the period have been geocoded to enable accurate computation of changes in key geospatial variables. Logistic regression is used to assess the independent effects of all variables in the model and estimate odds ratios of winning. Findings show that traveling across time zones is a disadvantage for all, particularly for teams traveling from west to east. While elevation and temperature changes are known to affect the results of other sports, they do not appear to be a significant factor in NCAA basketball tournament outcomes. The longstanding importance of home region advantage is also reinforced. The NCAA's adoption of the "pod format" for matching tournament seeds to sites succeeds in keeping higher seeds closer to home in the early rounds and in slightly improving their win percentages. However, the pod system, when introduced as a covariate in the model, does not exert a significant effect on tournament results or on the importance of geospatial variables in determining game outcomes.

Keywords: College Basketball, Geospatial Determinants, Circadian Rhythms, Time Zones, Elevation, Temperature, Distance Traveled, Home Court Advantage, NCAA Tournament, Bracketology, Pod Era

Introduction

Teams must travel through time and space to play against one another. Some travel thousands of miles and cross multiple time zones to compete. Some must endure significant changes in elevation and climate when they travel. And, of course, travel generally means playing farther from the coveted advantages of home region and a large local fan base. How these geospatial changes affect athletic performance has long been a subject of interest to all who have a stake in who wins and who loses in competitive sports. From players to fans and across the entire spectrum of the sports industry there is speculation about these effects; questions abound about what is fair or unfair, and what to do about it.

We see it in its extremes with Olympic athletes who travel weeks in advance to adjust to the host city time zone or train for months in advance to compete at high elevations such as Mexico City (Wilber, 2011). But for many competitive sports these adjustments are not an option. Professional and collegiate sports in the US, for example, maintain rigorous game schedules that can mean playing in Miami on one day and in Salt Lake City two days later, a 2,000 mile trip to a place that is 4,200 feet higher, two hours later, and 30 degrees colder. There is almost no time for players to adjust or for teams to prepare. How teams might be advantaged or disadvantaged by these geospatial movements, and combinations of these movements, is the focus of the present research.

NCAA Men's Basketball Tournament

The NCAA Men's Basketball Tournament provides an especially suitable context in which to examine the geospatial determinants of game outcomes. Each year, the tournament matches teams from all across the country over an intense three-week period in March and April, and event commonly known as "March Madness." Many teams travel long distances and to vastly different geospatial environments to compete. Because of limited budgets and short turnaround

times from one round to the next, players have little time to adjust, biologically or mentally, to their changing conditions. The Tournament Selection Committee is aware of some of the disadvantages of traveling through space and time and takes steps to incorporate these disadvantages into the seeding process (NCAA, 2013).

In recognizing the importance of travel and of home court advantage, the Committee has in recent years taken additional steps to reward better teams with closer-to-home venues, particularly during the earlier tournament rounds. In 2002 they adopted a “pod” format that enables groups of teams (usually 4 in a pod) to play the earlier rounds of the tournament in cities geographically distant from their regional brackets. For example, one of the cities in the 2013-14 South Region was Buffalo, NY, which is located in the northeastern US. The Four teams that made up one of the two Buffalo pods were Syracuse, Ohio State, Dayton and Western Michigan—all teams in relatively close proximity to Buffalo. Prior to 2002 the regional brackets were more “geographically pure” in that all tournament sites in a given region were actually located in that region of the country, requiring all teams not from that region to travel, often great distances, to play in cities in the region. As a result of the pod system higher seeds are now assured of travel relatively shorter distances on average, at least for the first two rounds of the tournament, and their winning percentages in these rounds have improved slightly as a result (Buikema 2012, McFall and Bronars 2012, 2010). In this paper we take a fresh look at the effects of the pod system in the context of the full range of geospatial variables in our results section below.

The Selection Committee’s primary goal is in evaluating each team and region fairly, and “getting the brackets right” is their charge (NCAA, 2013). While this is undoubtedly true and the Committee does the best with what it has, we also know that the job is as much art as it is science—perhaps more than it has to be. This is because the science, particularly our understanding of the geospatial determinants of tournament outcomes is not well established, as discussed below, and most assuredly our knowledge of these effects has not kept pace with our collective desire for the perfectly composed bracket, let alone with the event’s growing, multi-billion dollar revenue stream (Folger, 2011).

Previous Research

While there is much speculation and increasing scientific interest in the question of how geospatial changes affect game outcomes (e.g., Samuels, 2012), there is surprisingly little agreement on the extent or even the direction of the impacts (Carron et al, 2005; Smith et al, 2000). Particularly confounding are the intercorrelations among geospatial determinants. For example, crossing time zones (Winter et al, 2009; Smith et al, 1997; Steenland and Deddens, 1997), playing at higher elevations (Nassis, 2013; Pollard, 2006) or at colder temperatures are highly correlated with distance traveled and the advantages of a friendly fan base (Borghesi, 2007; Greer, 1983). We are familiar with no previous research on college or professional basketball that simultaneously models these variables to determine their independent and interactive effects on game outcomes. And whether these geospatial determinants have diminished in importance during the “pod era” of the NCAA tournament (starting in 2002), is another question worthy of empirical pursuit.

Most previous research on geospatial determinants has focused on how time zone changes disrupt circadian rhythms and how this “jet-lag” effect plays out in sports performance. For the interested reader, Reilly and Waterhouse (2009) have published a thorough review of the research literature on circadian rhythms and sports outcomes. They conclude from an extensive review of experimental and applied research that peak performance tends to occur in the late afternoon and early evening (4:00-8:00 pm) and suggest that travel across time zones that favors one team over another in terms of playing during this “biorhythmic sweet spot” can be seen in sports outcomes. This conclusion is consistent with the Smith et al. (1997) study of NFL Monday

Night Football games that found west coast teams traveling east to have a higher winning average than east coast teams traveling west. Arguably this is because Monday Night Football is played at 9:00 pm Eastern Time (6:00 pm Pacific Time), which is past the “circadian window” of maximum performance for eastern teams, but not so for their west coast opponents.

Steenland and Deddens (1997) also found that traveling east was relatively advantageous in their study of eight NBA seasons. They concluded that easterly travel effectively nullified the advantages of home-court advantage. Again, this may be a consequence of the fact that most NBA games are played at night, particularly those games played in eastern cities featuring west coast teams, the later scheduling being an important accommodation for west coast television viewers. Smith et al. (2000) found a similar but reduced effect of traveling east on NBA game outcomes, even after controlling for team strength.

The present research builds on past literature on geospatial determinants in two important ways. First, it moves it from the world of professional sports to the college level, notably to the NCAA Men’s Basketball Tournament where the daunting task of the Selection Committee has become the focus of increasing media and public scrutiny in recent years, along with the popularity of the tournament. Second, and more importantly, we are able to estimate the independent effects of four different dimensions of long distance travel on sports performance: playing farther from the fan base, crossing time zones, changes in elevation, and changes in temperature. In short, this research helps to “unpack” the constellation of variables that constitute home advantage (or “traveler’s disadvantage”).

Hypotheses

Our overall guiding hypothesis in this study is that geospatial change is disruptive and disadvantageous, and that the greater the change experienced in traveling to the game site, measured in terms of distance from home region, time zones crossed, higher elevation and change in temperature, the lower will be the odds of winning. We predict these results regardless of the direction of travel (east vs. west) or team strength. As a secondary hypothesis we posit that the greater advantages garnered by higher seeded teams since 2002 (start of the pod era) will result in less travel for high seeds and improved odds of winning, after controlling for other covariates.

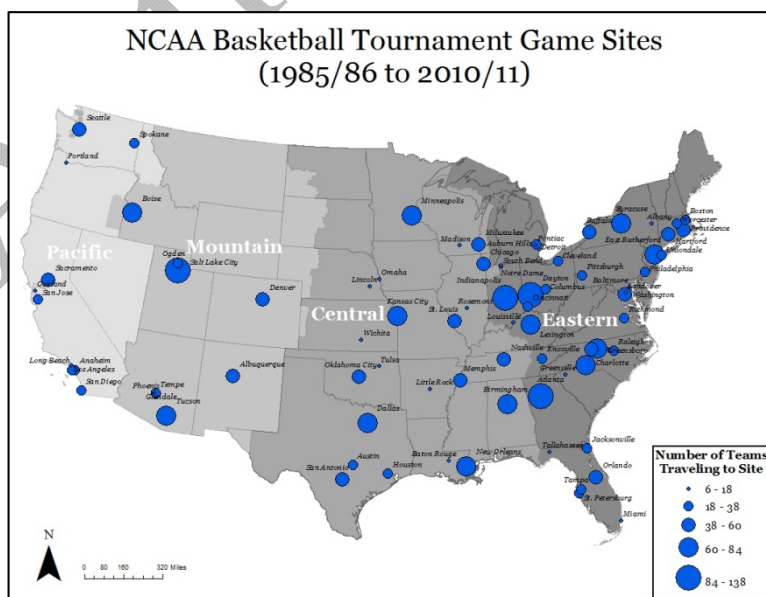


Figure 1

Data and Methods

Our analysis draws upon a data base of 3,296 individual team performances from all 1,648 NCAA Division I Men's Basketball Tournament games played over 26 years beginning with the 1985/86 season when the NCAA moved to the 64-team format and ending with the 2010/11 season. Represented in this data base are 278 teams from 38 college basketball conferences in the US and playing at 83 different tournament sites. These sites are mapped in Figure 1 with the symbol sizes increasing with the number of teams traveling to each site for tournament play across all rounds.

Each of the 3,296 team performances has been geocoded yielding latitude-longitude coordinates for both the home city and the tournament game site. The geo-coordinates enable the computation of distances traveled from home to the game site. Time zone, elevation above sea level and average temperature (in March) have also been added to the data base for each home city and tournament city. From these data, the changes in time zones, elevation and temperature experienced by each team are computed. Tournament game sites in the geographic information system (GIS) developed for this research are mapped in Figure 1, along with symbolic representation of their relative importance in terms of the number of teams traveling to each site over the 26-year study period. Time zones crossed are computed separately for those travelling east and those traveling west. These study variables are summarized below in Table 1.

Table 1: Study Variables Operationalization and Descriptive Statistics

<i>Variable</i>	<i>Description</i>	<i>N</i>	<i>Mean</i>	<i>Range</i>
<i>Game outcome</i>	Team won or lost (1 = won, 0 = lost)	3,296	0.50	0 - 1
<i>Time zones traveled east</i>	Number of time zones crossed traveling east	3,296	0.32	0 - 6
<i>Time zones traveled west</i>	Number of time zones crossed traveling west from team home to game site	3,296	0.45	0 - 3
<i>Change in elevation (ft)</i>	Difference in elevation (ft) between team site and game site	3,296	194.92	-7072 - 5182
<i>Change in temperature (°F)</i>	Difference in average March temperature (°F) between team site and game site	3,296	0.65	-34.3 - 38.1
<i>Distance traveled (mi)</i>	Miles traveled between team home and game site. Variable dummied (1=150+, 0=0-150) indicating team traveled outside of its "home region"	3,296	0.91	0 - 1
<i>Team strength</i>	Team seed in tournament (1=highest, 16=lowest)	3,296	6.70	1 - 16
<i>Tournament round</i>	Tournament round (0 = play-in round, 6 = championship round)	3,296	1.89	0 - 6
<i>Pod era game</i>	Game played during the "pod era" (1=2002 or later, 0=pre-2002)	3,296	0.35	0 - 1

Game outcomes are measured as a binary won-lost variable. Time zones traveled east and west range from 0 to 6 going east and from 0 to 3 going west. The difference is due to the occasional tournament participation by the University of Hawaii, which once during the past 26

years crossed six time zones for a tournament game. Average change in elevation is 195 feet, but ranges from -7,072 to 5,182 feet.

Distance travelled, beyond the corollary time zone effects, is found to be a close proxy for “home region advantage” and as such is converted to a binary dummy variable, where 0 = 0-150 miles from home city and 1 = 150+ miles traveled, indicating that the team traveled outside of its favored “home region.” Various distance thresholds were tested for goodness of fit and it was found that the distance effect waned sharply after approximately 150 miles, or about a 3-hour drive from the home town. A second reason for the drop-off in the distance effect is because teams tend to travel by airplane for tournament trips greater than 3-4 hour’s drive; flying reduces the travel time so significantly that it seems to matter little whether the site is 300 or 1,000 or more miles away.

Results and Discussion

Figure 2 compares winning percentages by the number of time zones traveled for teams traveling east and for those traveling west, controlling for no other variables. While there is a slight decline in outcomes for teams going west, the decline is precipitous for those tournament teams that travel east, dropping to below 38% for teams crossing two or more time zones.

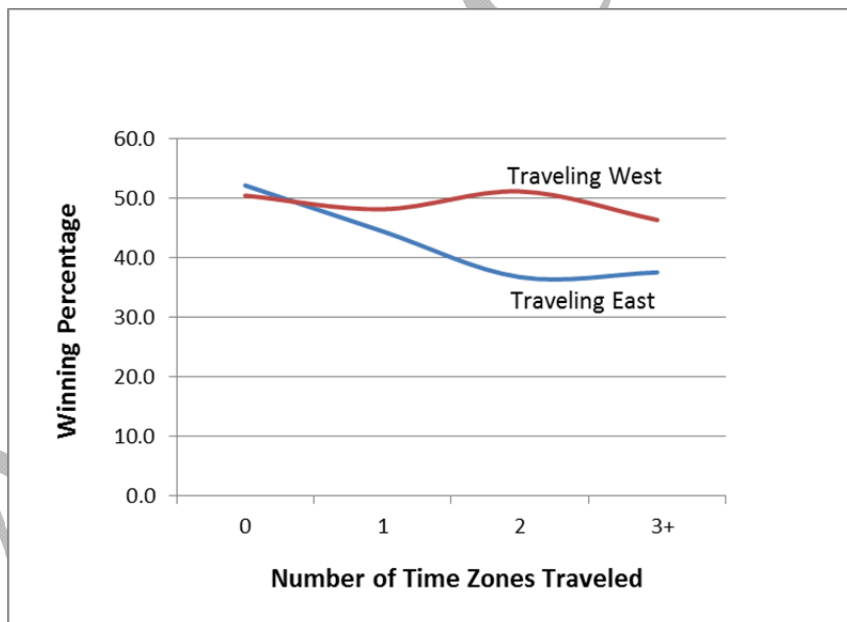


Figure 2: Winning Percentage by Number of Time Zones Traveled

To help visualize these movements geographically Figure 3 maps direction of travel, number of time zones crossed, volume of team travel and win percentage for the instances of team travel from one time zone to the next. Win percentages are generally in the lower categories for teams traveling east and higher for teams traveling west. For example, teams traveling from the Mountain time zone win only 40.9% of their games played in the Eastern time zone compared to a 55.0% winning rate for teams traveling in the opposite direction, from the Eastern to Mountain time zones. A second observation taken from this map is that the volume of travel is greater for

games played just one time zone away compared to those requiring teams to cross two or three time zones to play. This can be explained by the Selection Committee’s commitment to keeping teams, especially higher seeds, close to home whenever possible.

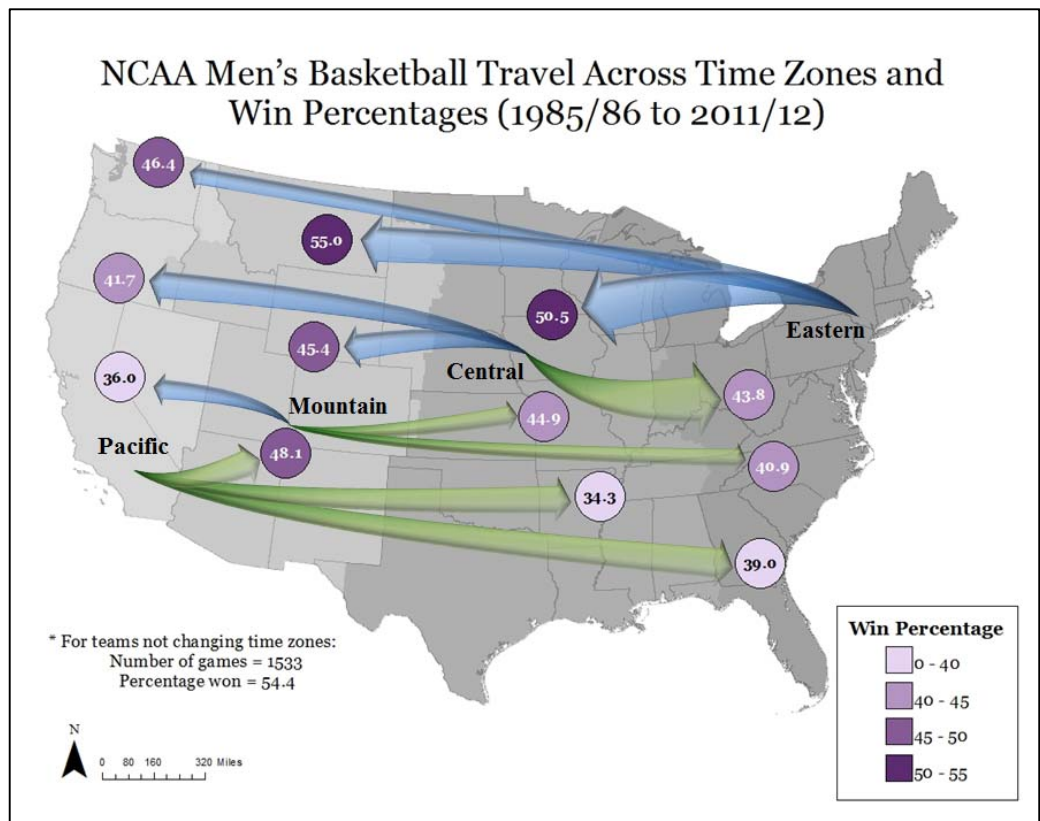


Figure 3

Examining how game outcomes relate to other key study variables, Table 2 reports the results of an analysis of variance (ANOVA) which compares means across the four geospatial variables for teams that won versus teams that lost, then tests for the statistical significance of those differences. The findings show, at a bivariate level, that on average there is little difference in temperature and elevation change by game outcome. However, distance travelled and strength of team as reflected in mean tournament seed are both significantly associated with game results, with higher seeded teams and those playing close to home being the more likely winners. In NCAA Tournament play, higher seeded teams are more often protected against the disadvantages of long distance travel, and this is expected to be even truer today with the adoption of the pod bracket system in 2002.

Table 2: Comparison of Means (ANOVA) by Game Outcomes

Predictor Variable	Game Outcome			F	Sig.
	Won	Lost	Total		
Mean Difference in Temperature (°F)	0.82	0.48	0.65	0.71	0.40
Mean Difference in Elevation (ft)	221.20	168.60	195.00	1.02	0.31

<i>Mean Team Distance (1= 150+ miles)</i>	0.88	0.94	0.91	33.93	0.00**
<i>Mean Tournament Seed</i>	4.74	8.65	6.70	734.00	0.00**

** Significant at the 0.01 level (2-tailed). * Significant at the 0.05 level (2-tailed).

To examine the impact of the “pod format” adopted by the NCAA Selection Committee in 2002 that favors higher seeded teams with closer-to-home tournament sites, we plotted the distance travelled by two seed groups, high seeds (1-4) and lower seeds (5-16), over the 26 year period. Only data for the rounds of 64 and 32 are plotted as the pod system is primarily targeted to these two rounds. Together they account for 76% of all tournament games played. Figure 4 shows that the higher seeds have historically had relatively shorter travel distances than lower seeds in most years (average 128 fewer miles). The introduction of the pod format in 2002 has resulted in even greater separation of the two seed groups. The top four seeds now travel 435 fewer miles on average compared to the lower seed group. One also finds that the change in format has led to a slightly higher winning percentage among the top seeds in the first two rounds of tournament play, rising from 79.1 percent before 2002 to 80.5 percent since that time.

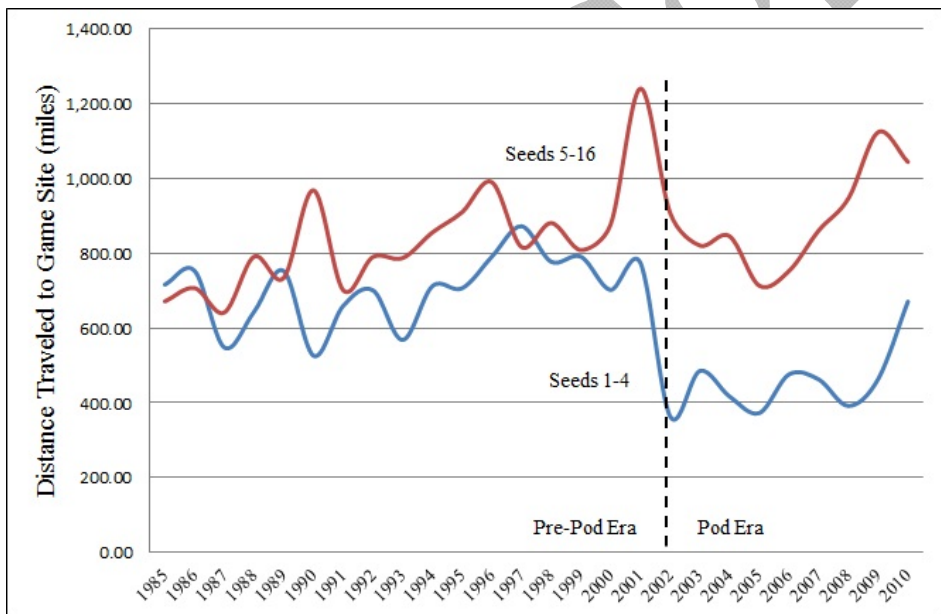


Figure 4: Distance Travelled by Year and Seed Group (NCAA Rounds 1 & 2)

The final stage of our analysis combines all of our study variables in a logistic regression model. Logistic regression is used in this case as a technique designed to estimate the effects of predictor variables on a binary dependent variable, that being game outcomes (won – lost). In this case, the independent effects of each of the geospatial variables on game outcomes are estimated while controlling for team strength (seed) and the effects of the pod era changes in site selection. From the logistic regression model results presented in Table 3 it is observed that traveling east across time zones significantly reduces the odds of winning (odds ratio of .861), or 13.9% lower than for those not traveling east. Traveling west also reduces the odds of winning but not significantly. These results reinforce those reported earlier in Figure 1, enabling us to draw the same conclusions about the difficulties of traveling across time zones, yet with even

greater certainty as they are shown to hold up when controlling for the effects of other geospatial variables and for strength of team.

What is most intriguing is that these findings run counter those found in previous research on NBA games and NFL Monday Night Football, both of which reported an advantage to easterly travel. We believe the difference lies in the time of day that the games are played. Monday Night Football and most NBA games are played at night, sometimes very late at night in the eastern time zone (9:00 – 12:00 midnight), enabling west coast teams to benefit by playing during the high performance hours of their circadian rhythms (4:00 – 8:00 pm). NCAA tournament basketball, however, is played throughout the day, particularly for the early rounds, when over three-fourths of the games are played. The majority of these games are played in the afternoon, enabling eastern teams to compete during or closer to the hours of their biorhythmic sweet spot than teams from farther west. As a consequence the “jet lag” effect is minimized for those teams traveling from east to west. In other words, the time of day that NCAA Tournament games are scheduled can favor one team over another when they originate from different time zones.

Table 3: Logistic Regression Estimating Game Outcomes by Geo-Spatial Changes, Tournament Seed, Round and Pod Era Brackets

<i>Variable in Equation</i>	<i>B</i>	<i>S.E.</i>	<i>Wald</i>	<i>Sig.</i>	<i>Odds Ratio Exp (B)</i>
<i>Time zones traveled east</i>	-.150	.064	5.486	0.019*	0.861
<i>Time zones traveled west</i>	-.065	.057	1.283	0.257	0.937
<i>Elevation change (ft)</i>	.000	.000	1.253	0.263	1.000
<i>Temperature change (°F)</i>	.003	.004	.483	0.487	1.003
<i>Distance traveled dummy (>150+ mi = 1)</i>	-.440	.154	8.181	0.004**	0.644
<i>Strength of team</i>	-.279	.011	599.792	0.000**	0.757
<i>Tournament Round</i>	-.483	.038	164.359	0.000**	0.617
<i>Pod era game (2002+)</i>	.002	.084	.001	0.982	1.002
<i>Constant</i>	3.237	.194	278.128	0.000**	25.448

** Significant at the 0.01 level (2-tailed). * Significant at the 0.05 level (2-tailed).

As observed earlier in the one-way analysis of variance, elevation and temperature change do not affect NCAA tournament game outcomes, even when team strength and other model variables are controlled in the logistic regression model. It was surmised that traveling to a higher elevation would reduce the odds of winning, as has been witnessed in other sports, yet this hypothesis is not borne out in the data. By contrast, home region advantage, measured as a distance travelled dummy, does exert the expected strong effect on game outcomes. Traveling more than 150 miles from home reduces the odds of winning to 0.664, or 33.6% less than for those who play in their home regions, independent of team strength.

Finally, to test the effect of the pod format (adopted in 2002) on how successfully other variables in the model predict outcomes, we introduce the pod era variable as a final step in the model. As shown in Table 3, controlling for the pod era years does not increase the effectiveness of the model in terms of variance explained or predictive capacity (percentage correct), nor do the odds ratios of any of the other variables in the equation change in appreciable measure.

Conclusion

This research helps to clarify longstanding speculation and debate about how travel across space and time can affect Men's NCAA basketball game outcomes. Using a complete and fully geocoded data base including all NCAA Tournament games during the 26 year period from 1985/86 to 2010/11, we have managed to “unpack” the combination of geospatial changes that teams endure as they travel from their home cities to tournament sites to compete against one another. Guided by previous research on time zone changes and the effects on circadian rhythms (jet lag effects) easterly and westerly travel were analyzed separately. We found that traveling east reduces the odds of winning more than does traveling west, and this finding holds when controlling for strength of team, home region advantage and other covariates. Traveling longer distances (>150 miles) also has a dramatic negative effect on game outcomes as it corresponds directly to the erosion of fan support which is invariably stronger and friendlier closer to home. Elevation and temperature changes can be dramatic for many teams, especially as they travel west and north to sites such as Salt Lake City and Minneapolis, but we find that neither of these geospatial changes exerts a measurable effect on Tournament game outcomes.

While adoption of the pod format has effectively reduced distances traveled by higher seeds in the early rounds of the Tournament, and has slightly improved win percentages in these rounds, we did not find it to exert a significant broader effect on tournament results or on the importance of geospatial variables in determining game outcomes.

Our results hold important practical implications for sports scheduling and seeding, as well as considerations for pre-game travel and preparation, not to mention bracketology, the emergent US national pastime in the month of March. It is imperative that coaches and players be aware of what long distance travel means to performance on the court and how its negative effects can be mitigated. Additionally, how these geospatial determinants are taken into consideration by the NCAA Tournament Selection Committee in establishing brackets is no small matter. It is complicated enough just identifying the 64 teams, seeding them and placing them in regional brackets. Where the findings from this research will fit into the process may be more nuanced, particularly in terms of which direction teams must travel and at what time of day they play. There is some flexibility in these aspects of the process and they will provide the Committee additional levers to mitigate the disadvantages of long distance travel (especially easterly travel) that some teams inevitably face. We expect that insights presented in this discussion will help to make the Committee's job a notch fairer, but likely not any easier.

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